

find Cheat Sheet

Search folders for files by name, type, size, age, owner or permissions, then act on them.

Using it

```
find [where] [tests] [action] ← no where = current folder · no action = print each path
find /var/log -name '*.log' -mtime +30 ← .log files under /var/log changed more than 30 days ago
```

Tests (what to match)

Test	Matches	Test	Matches
-name '*.log'	File name matches (case-sensitive)	-mtime -7	Modified in the last 7 days
-iname '*.jpg'	File name, ignoring case	-mmin -60	Modified in the last 60 minutes
-type f	Regular files (d = folders, l = links)	-newer ref	Modified more recently than file ref
-size +100M	Bigger than 100 MB (k M G)	-user mark	Owned by user mark
-empty	Empty files and empty folders	-perm -u+x	Owner can execute it
-path '*/tmp/*'	The whole path matches	-maxdepth 2	Goes at most 2 folders deep

Reading the numbers: + and –

Written as	Means	Example
+n	More than n	-size +1G = bigger than 1 GB · -mtime +30 = older than 30 days
-n	Less than n	-size -10k = smaller than 10 KB · -mtime -1 = changed in the last day
n	Exactly n	-mtime 0 = changed within the last 24 hours

-mtime counts whole days and rounds down, so **-mtime +1** means at least **2** days old.

Combining tests

Write	Means	Write	Means
-name a -type f	And (tests side by side)	\(... \)	Groups tests (spaces needed inside)
-o	Or: -name a -o -name b	-prune	Skips a folder (see recipes)
! or -not	Not: ! -name '*.bak'		

Actions (what to do with each match)

Action	Does	Action	Does
-print	Prints the path (the default)	-exec cmd {} \;	Runs cmd once per match ({} = the path)
-ls	Prints size, owner, date, permissions	-exec cmd {} +	Runs cmd once with many paths (faster)
-delete	Deletes the match. Put it last	-ok cmd {} \;	Like -exec , but asks first

Handy recipes

Command	Does
find / -type f -size +1G 2>/dev/null	Files over 1 GB; hides "Permission denied"
find . -name '*.tmp' -mtime +7 -delete	Deletes .tmp files older than a week
find . -type f -name '*.py' -exec grep -l 'TODO' {} +	Lists Python files that contain TODO
find . -name node_modules -prune -o -name '*.js' -print	Finds .js files, skipping node_modules
find . -type d -empty	Lists empty folders
find /etc -type f -mmin -30	Config files changed in the last 30 min
find . -type f -exec du -h {} + sort -rh head	Shows the 10 biggest files

Remember: quote name patterns (**'*.log'**) so the shell doesn't expand them first. Before using **-delete**, run the same command without it and check the list.